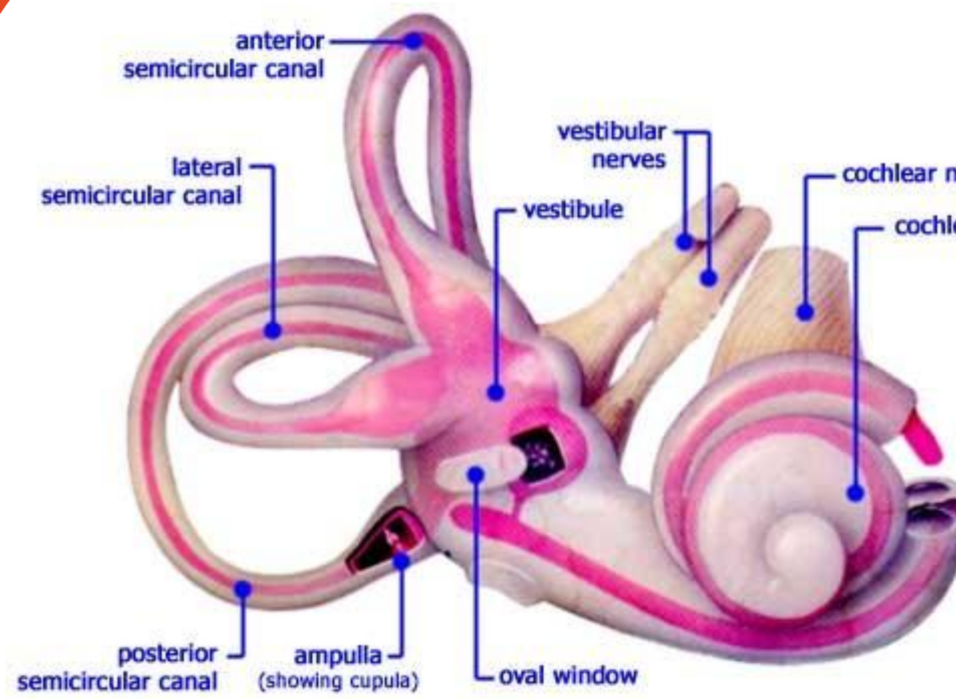
The Vestibular System



The Vestibular System

1. The anatomy of the inner year
2. The vestibular apparatus
3. The vestibular receptors
4. The transduction mechanism
5. The central vestibular pathways

Learning Objectives

After studying these material you will be able to:

1. Discuss the function of the vestibular system
2. Describe the anatomy and histology of the vestibular apparatus
3. Explain the location and composition of the endolymph and perilymph
4. Describe the type of head movement detected by the different receptors:
   - Angular acceleration – semi-circular canals
   - Linear acceleration – utricle and saccule
5. Explain the localization, morphology and function of the vestibular receptors: the hair cells and discuss the transduction mechanism in the vestibular system
6. Explain the central vestibular pathways:
   - The vestibular nuclei and their connections with cerebellum, spinal cord, and eye movement centers
   - The central pathways to the cerebral cortex
7. Explain the function and physiology of the vestibulo-ocular reflex
8. Describe the caloric testing
9. Apply the acquired knowledge to recognize peripheral and central disorders of the vestibular system

The Vestibular System

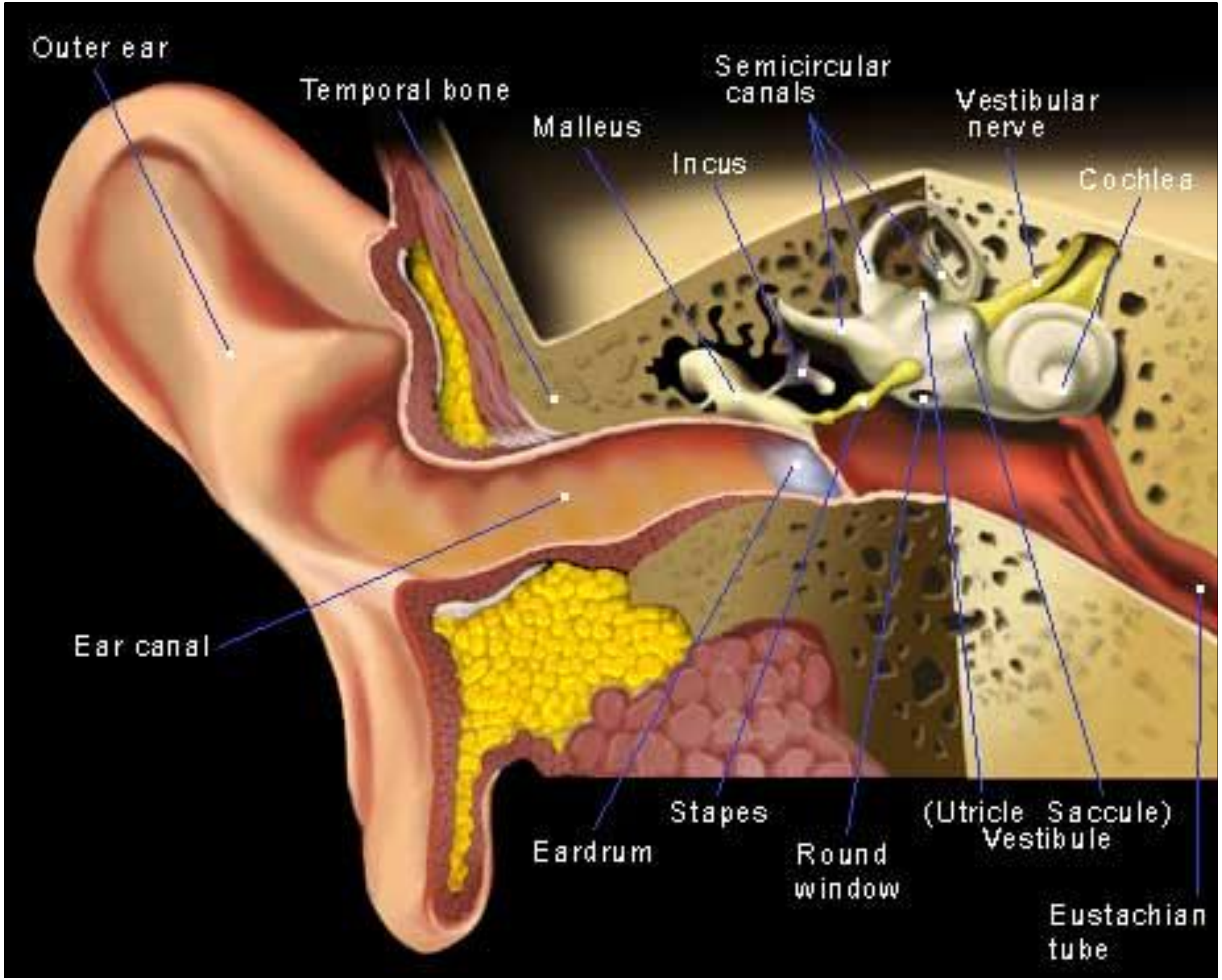
OBJ. # 1

Function:

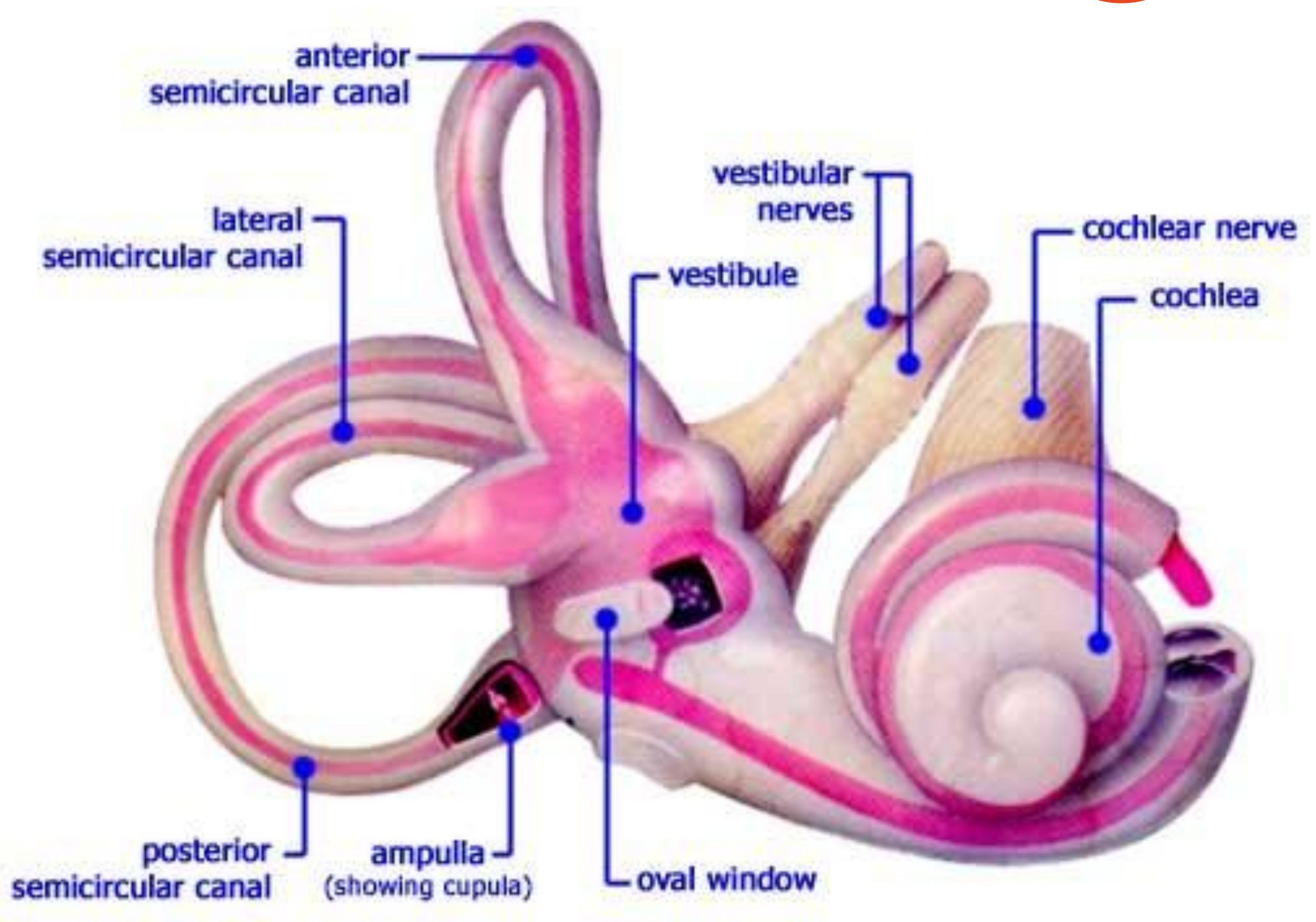
Provides information about head and body position in the space and allows for rapid compensatory responses to both self-induced and externally generated forces

→ **Postural reflexes and eye movements**

OBJ. # 2


Outer ear
Temporal bone
Malleus
Semicircular canals
Incus
Vestibular nerve
Cochlea
Ear canal
Eardrum
Stapes
Round window
(Utricle Saccule)
Vestibule
Eustachian tube

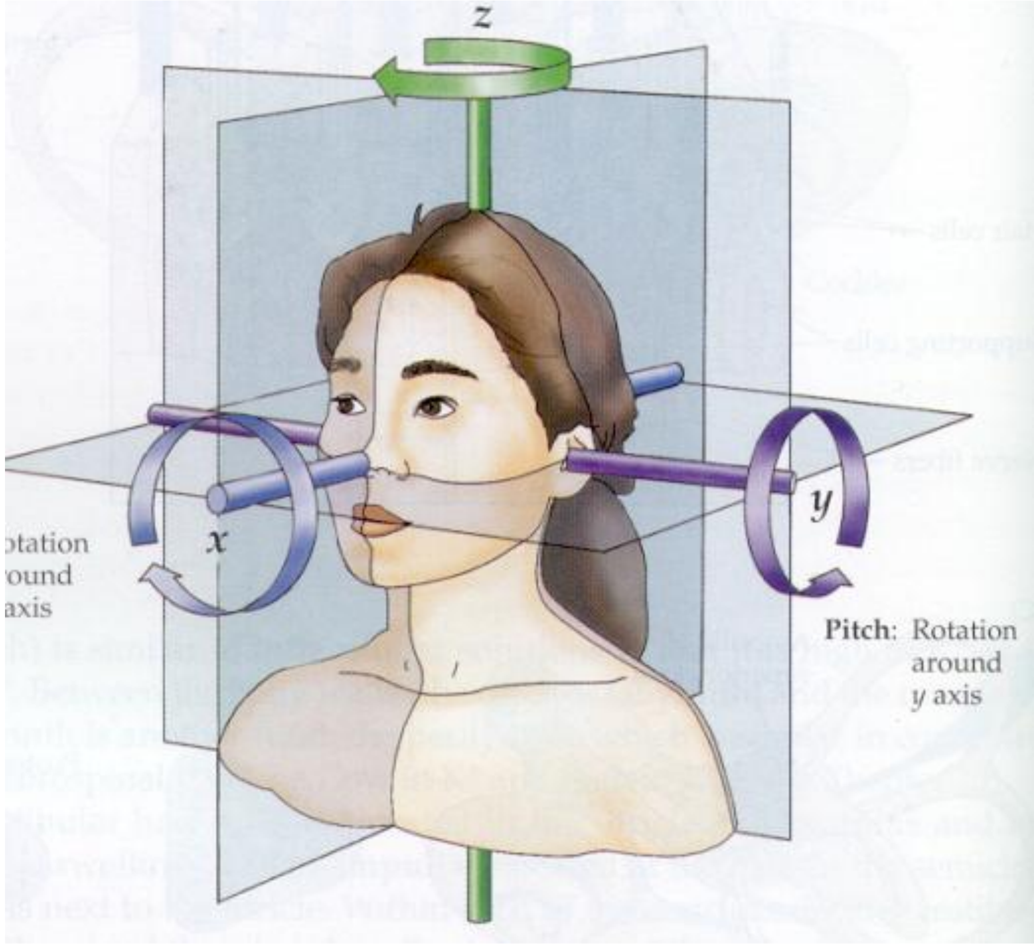
The Inner Ear



OBJ. # 2 & 3

Head Movements Detected By The Vestibular Receptors

OBJ. # 4



Vestibular receptors - 5 different locations

At the crista ampullaris of the semi-circular canals:

- Anterior or superior
- Horizontal or lateral
- Posterior

Sense angular acceleration of the head (head rotation)

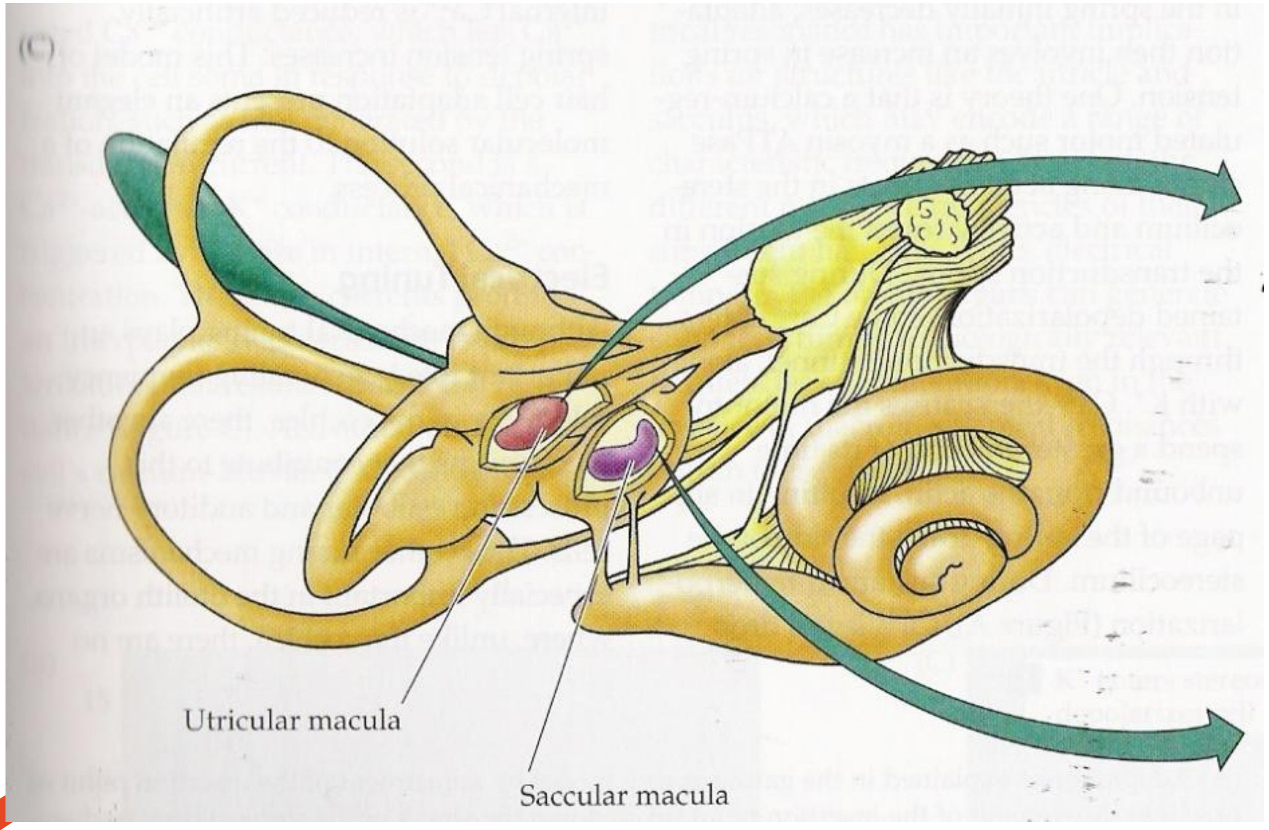
- The macula of the utricle and saccule

Sense linear acceleration (translation),
Static head position and head tilt relative to gravity

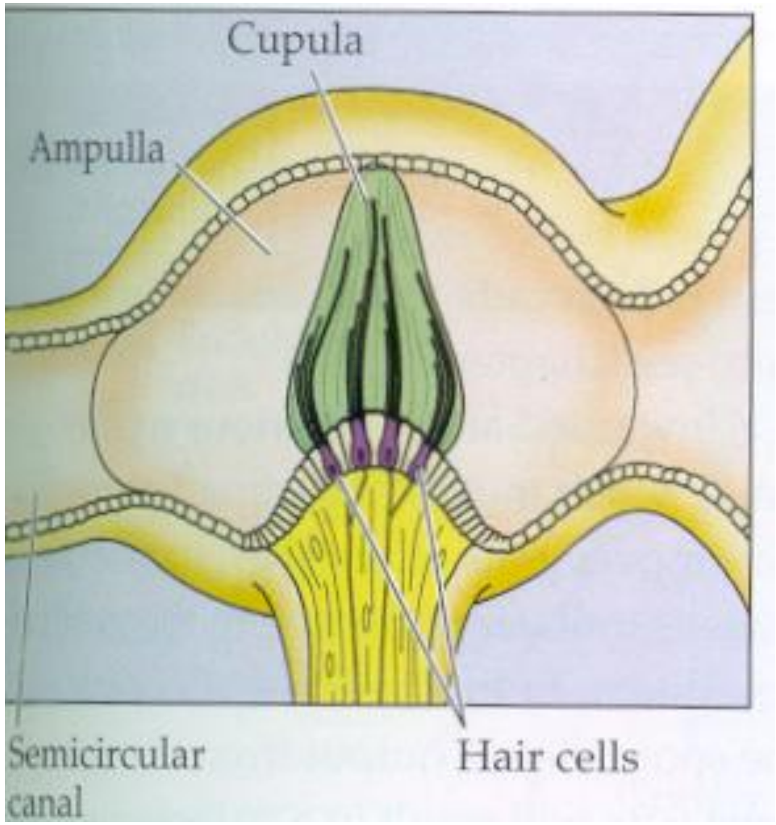
The Vestibular Receptors

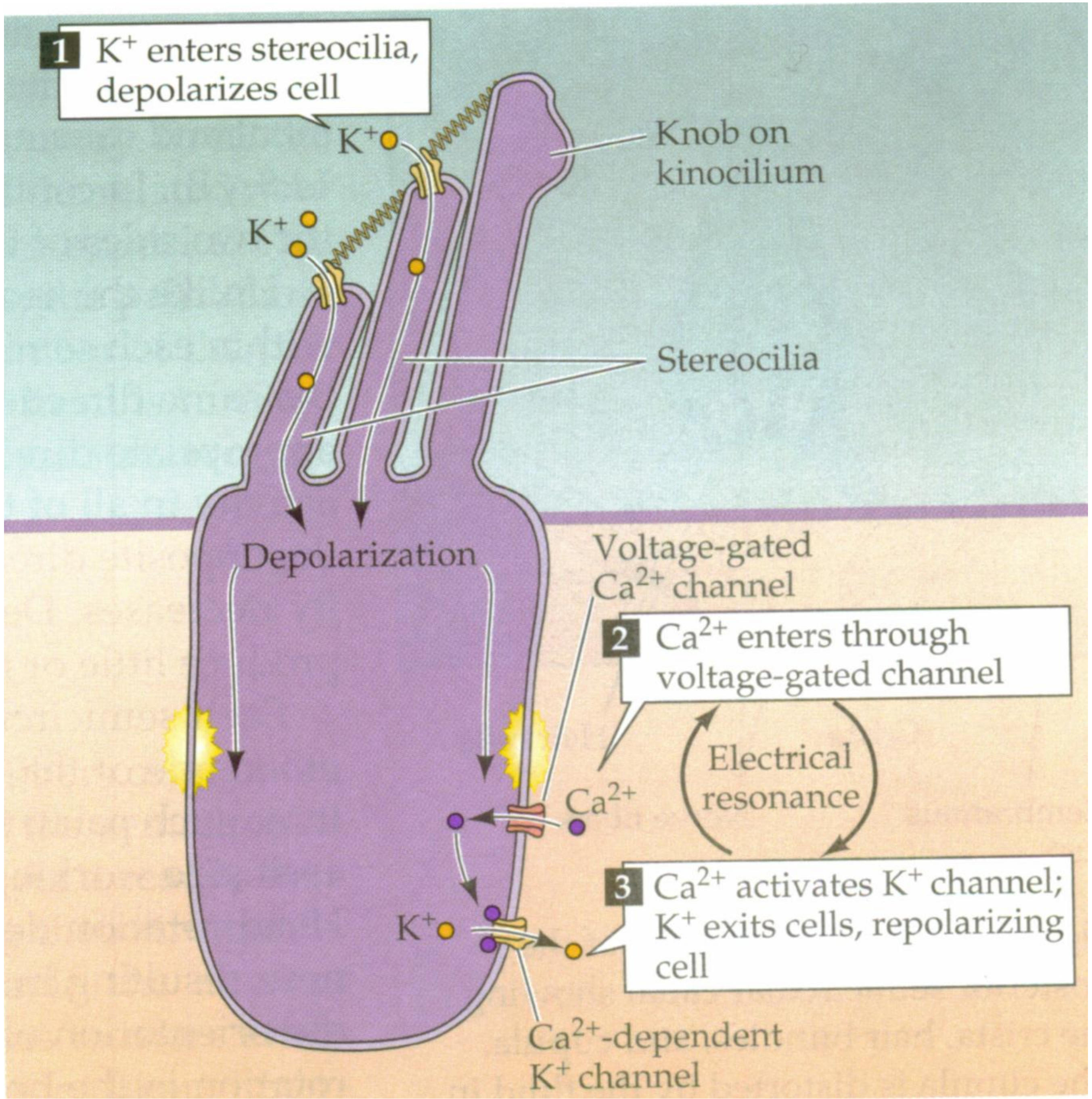
OBJ. # 5

Utricle and saccule



Semi-circular canals





The Hair Cells And The Transduction Mechanism

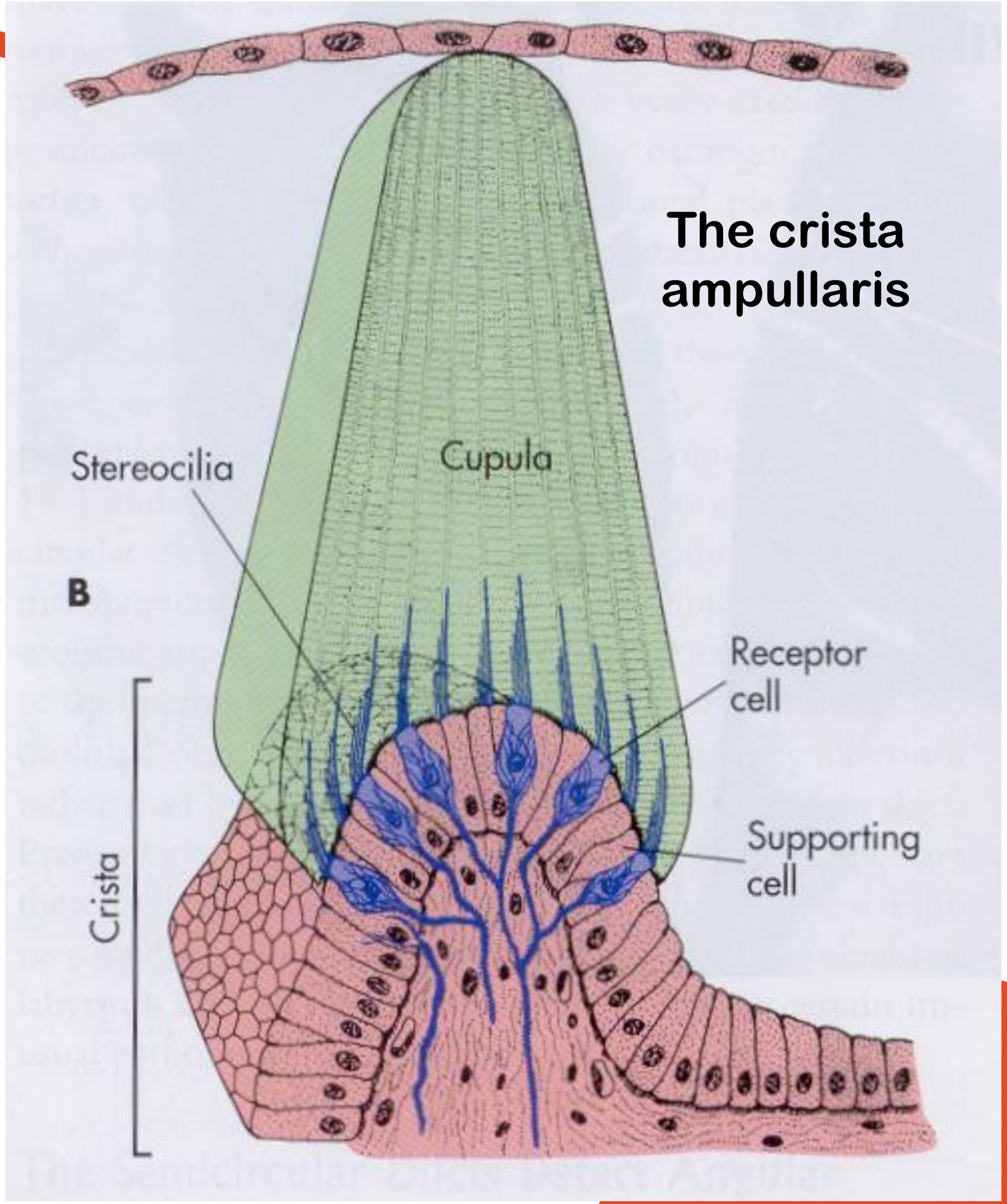
OBJ. # 5

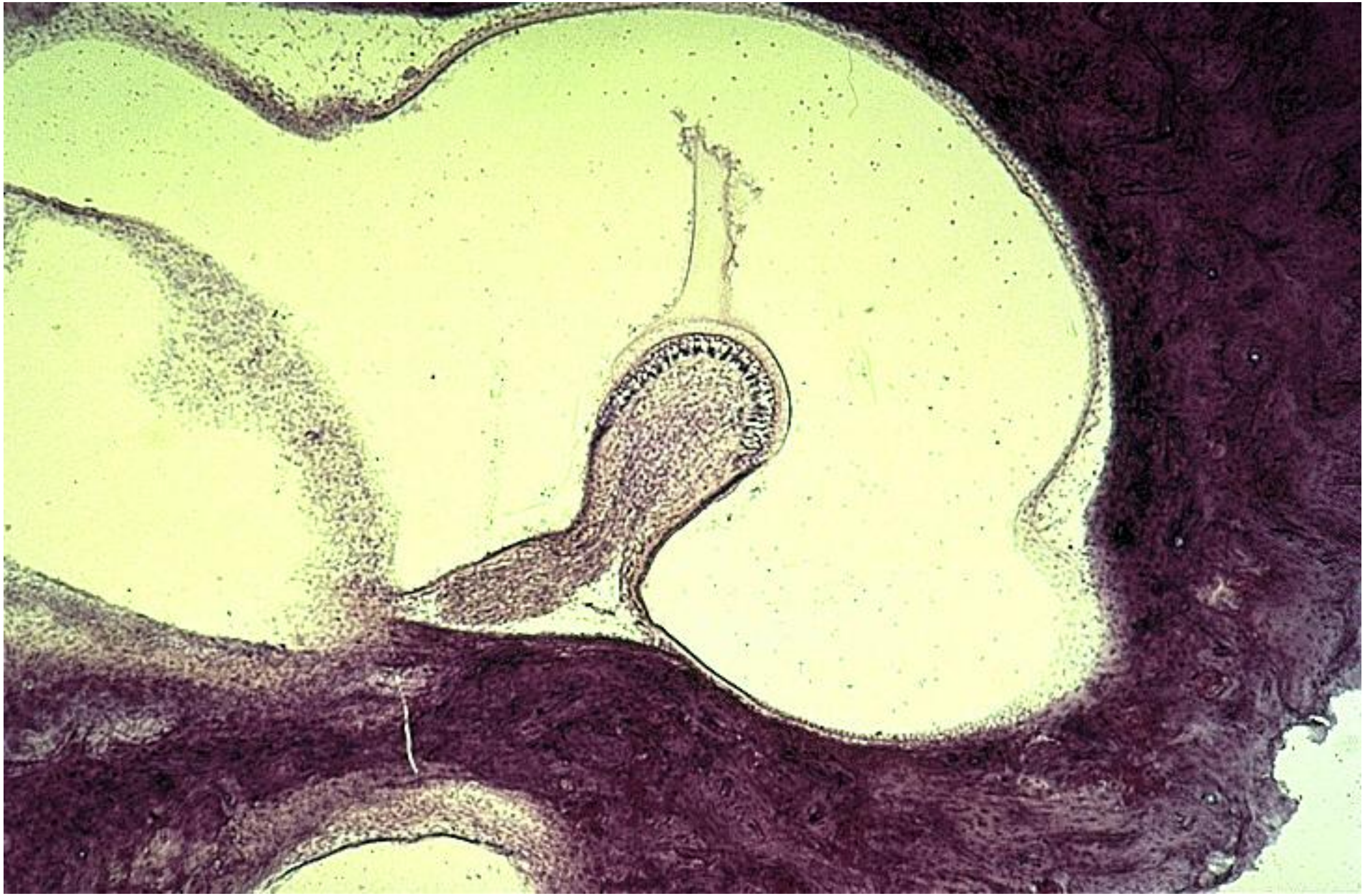
The Crista Ampullaris in the Semi-circular Canals

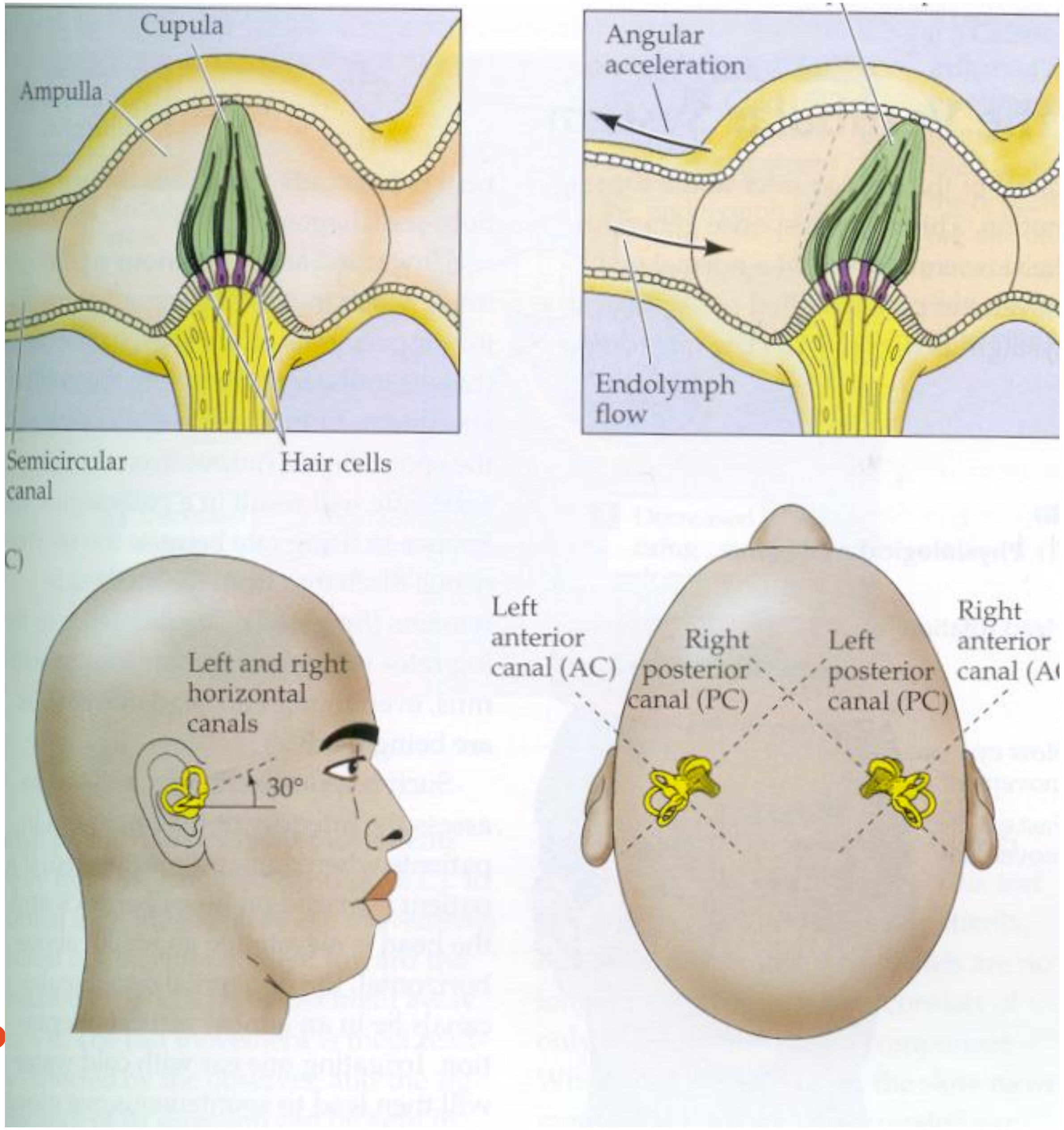
The hair cells of the semicircular canals are located in specialized dilations called ampullae

The sensory epithelium in each ampulla is called the crista, it contains the hair cells

OBJ. # 5



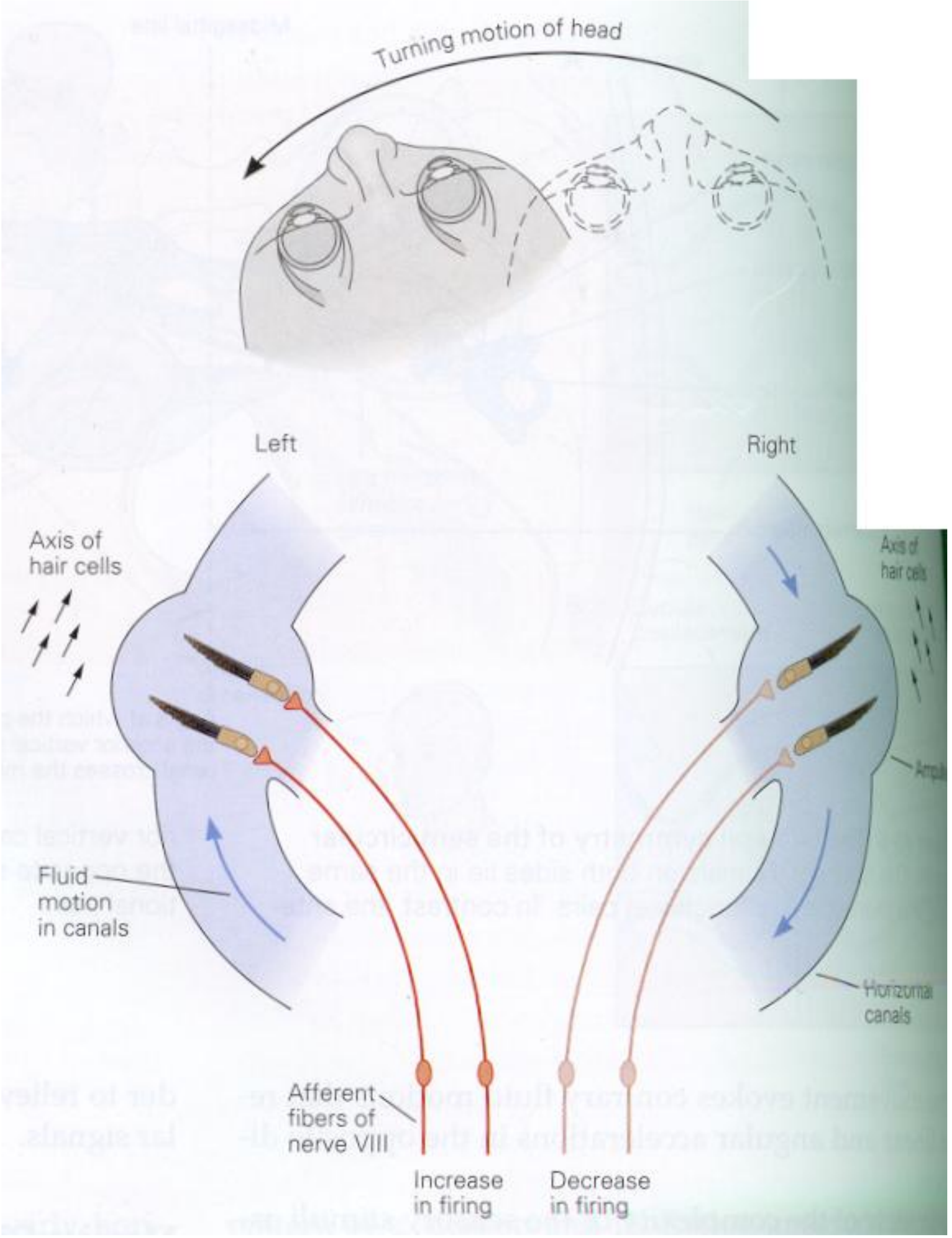




The Semi-circular Canals

How the Semi-circular canals detect movement?

OBJ. # 5



Angular acceleration of the head to the left produces movement of the endolymph to the right. Deflection of the stereocilia towards the utricle depolarizes the hair cells on the left. The opposite occurs on the right side with hyperpolarization of the hair cells

OBJ. # 5

The Vestibular Receptors

The hair cell bundles in each vestibular organ have specific orientations

The utricle and saccule contain a sensory epithelium called the macula

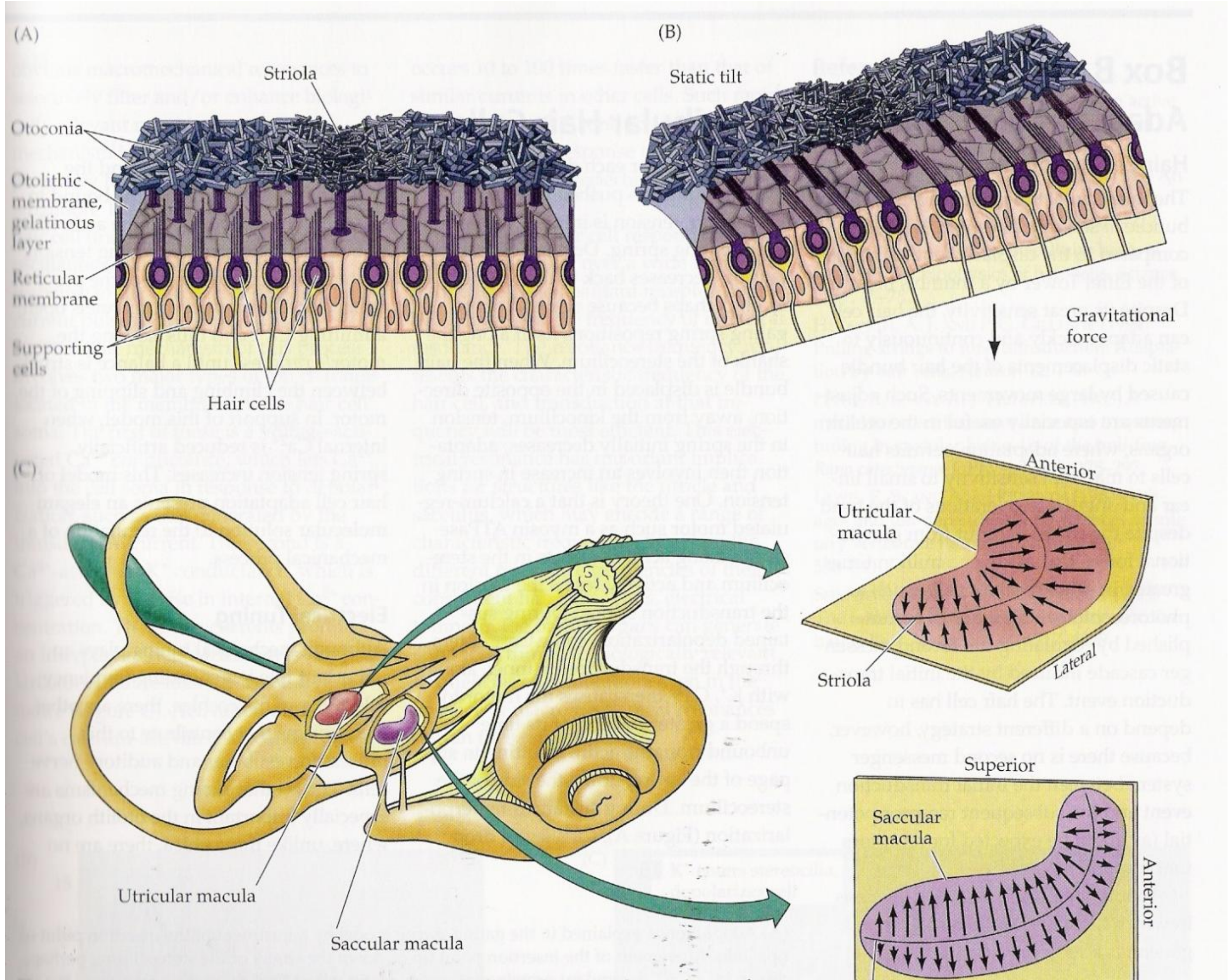
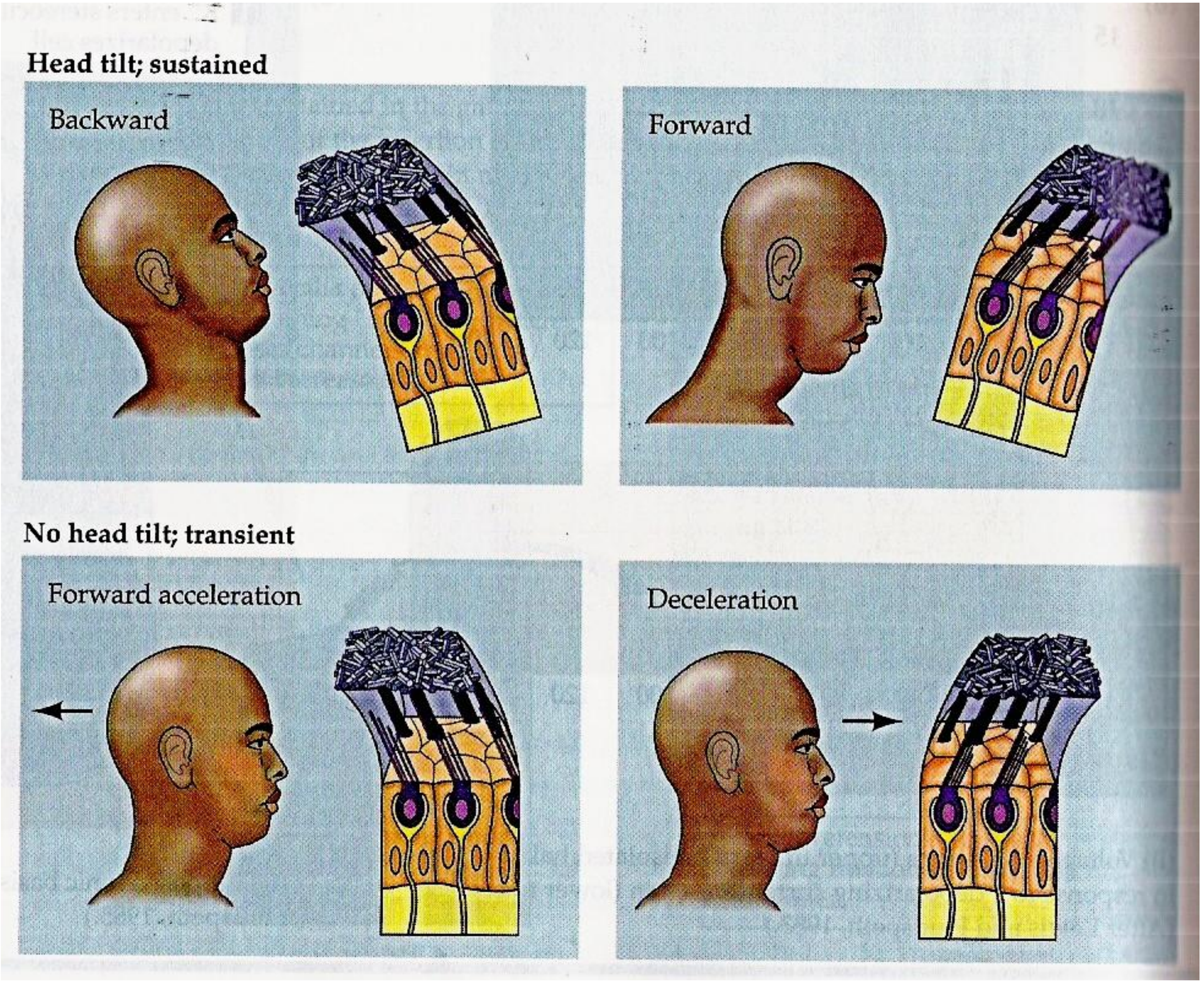


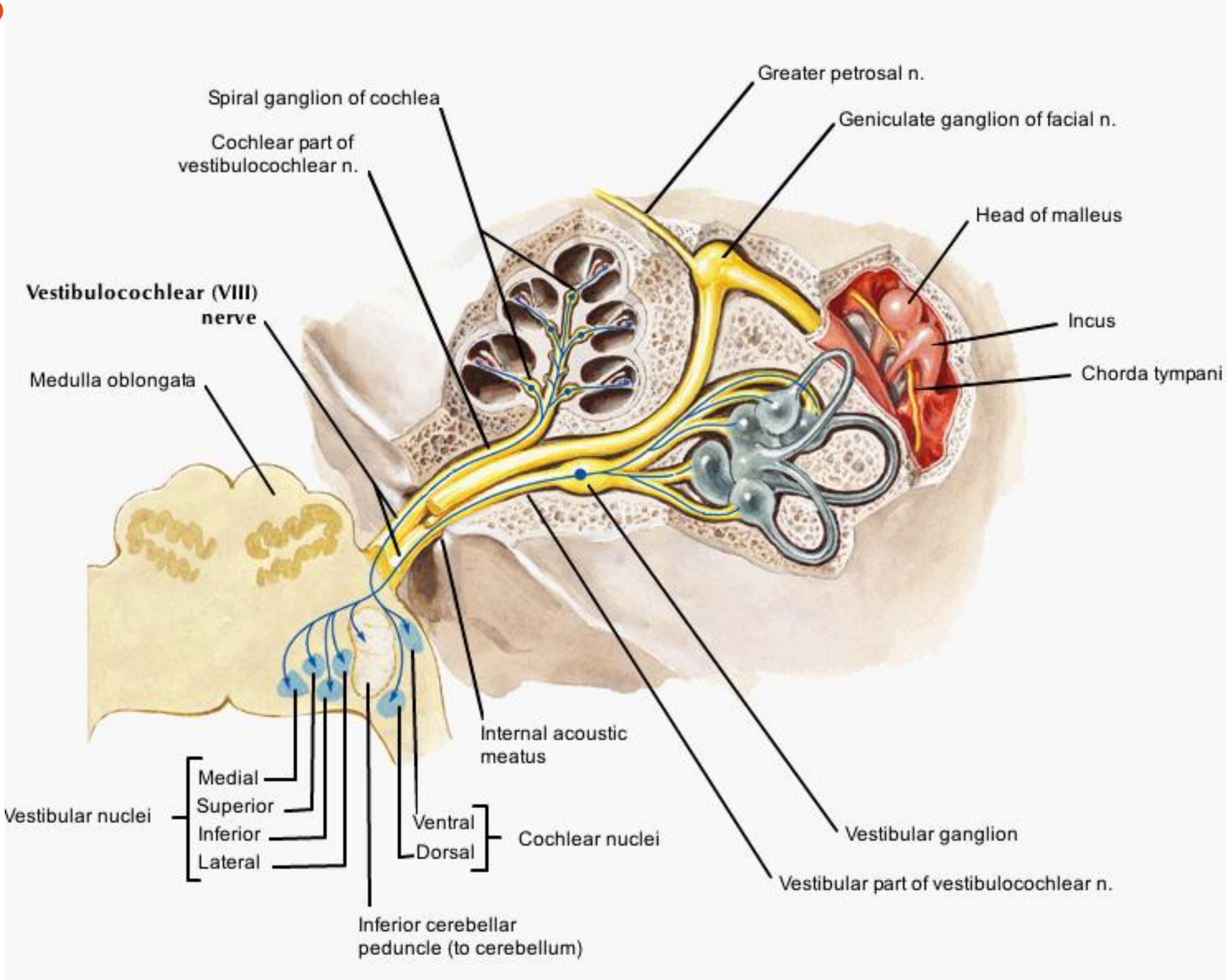
Figure 13.4 Morphological polarization of hair cells in the utricular and saccular maculae. (A) Cross section of the utricular macula showing hair bundles projecting into the gelatinous layer when the head is level. (B) Cross section of the utricular

Linear acceleration

The Vestibular Receptors

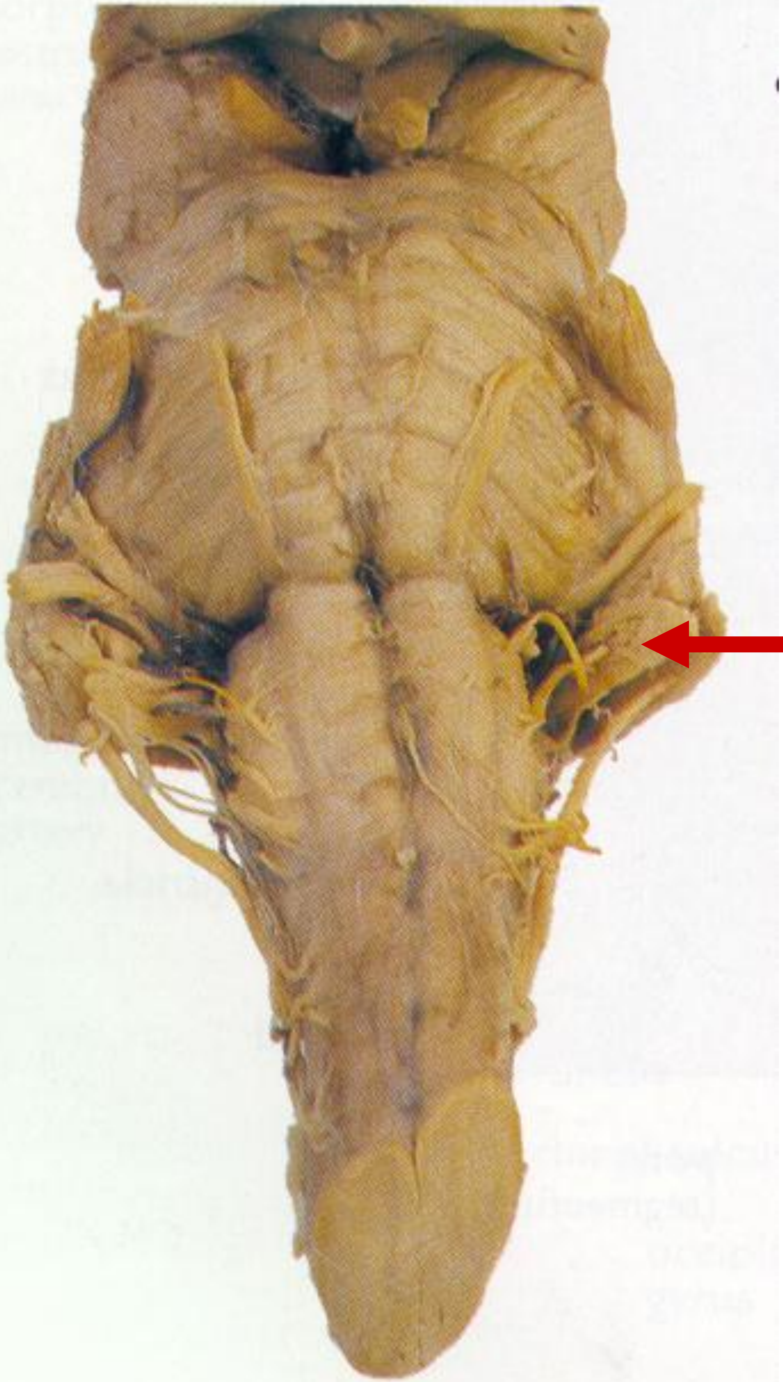


OBJ. # 6



The Vestibular Pathways

The vestibular receptors are contacted by the primary afferent fiber



CN VIII
Vestibulocochlear
at pontocerebellar
angle

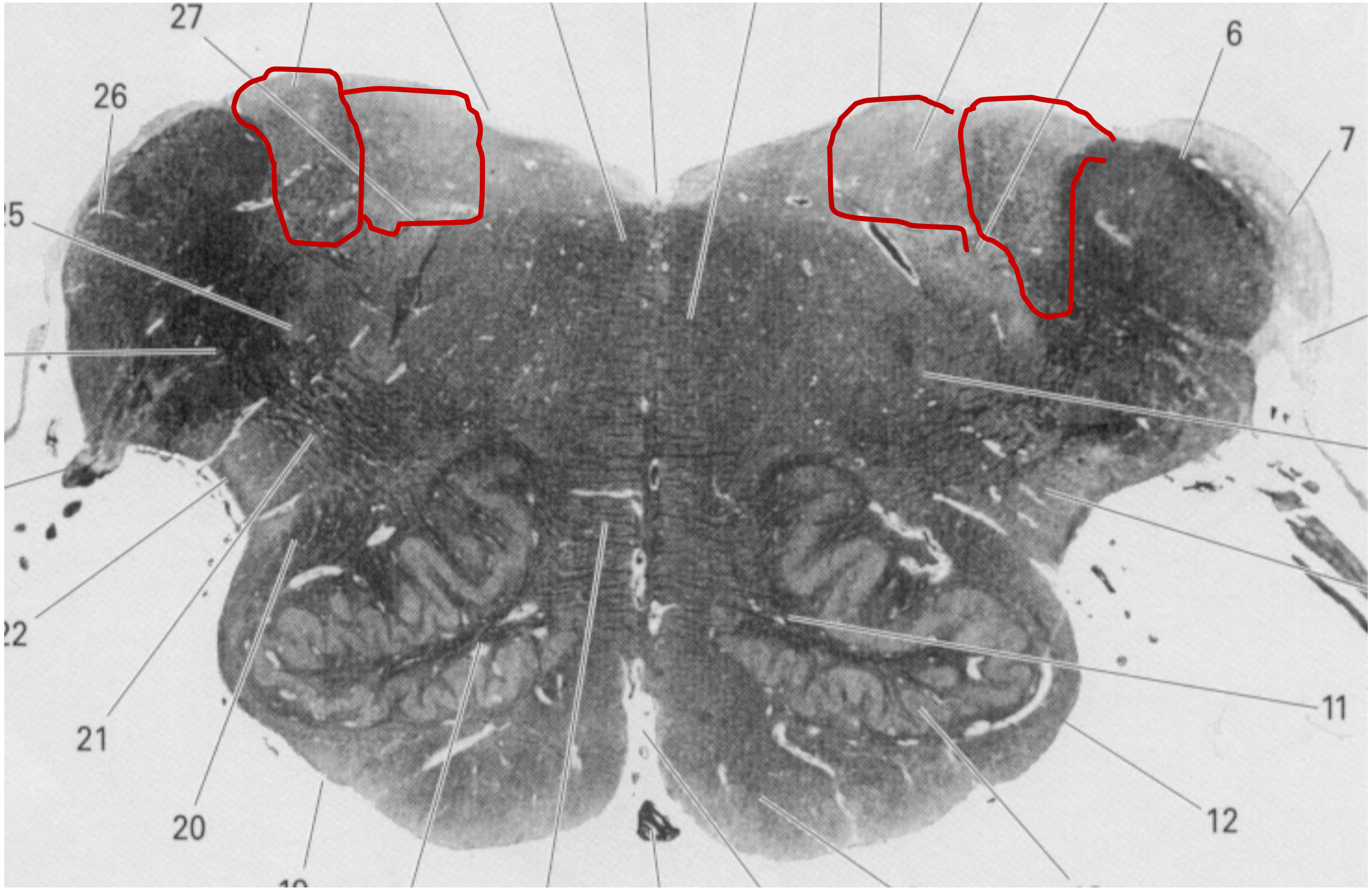
Ventral Surface Of The Brainstem

OBJ. # 6

Rostral Medulla

Location of 2 Vestibular nuclei on the floor of the IV ventricle

OBJ. # 6



The Central Vestibular Pathways

OBJ. # 6

(Ant. Inf. Cereb. artery)

1st Neuron
Scarpa's ganglion

Juxtarestiform
Body Ipsilaterally

Vestibulocerebellum
- Eye Movements
- Head Movement
- Posture

Primary fibers

Secondary fibers

Modulatory Inputs

Reticular
Formation

Spinovestibular
Fibers (Propioception
all levels of spinal cord

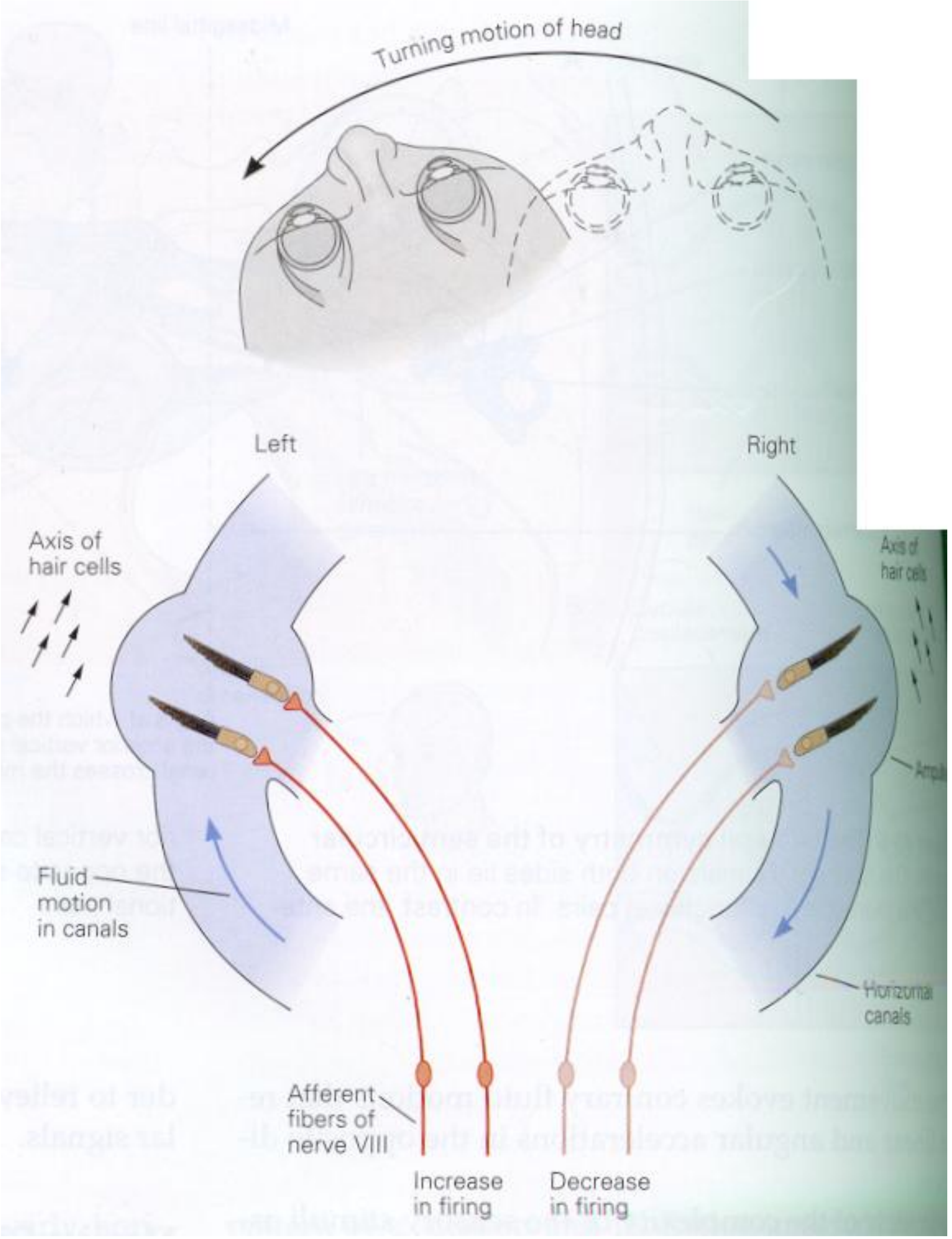
2nd
Neuron
Vestibular
Nuclei

Vestibulospinal fibers
- **Lateral VS Tract**
  All levels of spinal cord, ipsilateral to A, G motor neurons.
  - Postural control
  - Excitatory input on extensors
- **Medial VS Tract**
  Stimulation to neck muscles bilaterally MLF to cervical spinal cord
  - Control of head and neck position

Gaze Centers /
Nuclei of CN III,
IV, VI MLF
(conjugate gaze)

Vestibulo-thalamo-cortical Fibers
- Cognitive perception of motion and spatial orientation is:
  - Vestibular
  - Visual
  - Somatosensory
- Vestibular fibers bilaterally to VPL, and other ventroposterior tier nuclei

Cortex: = Posterior parietal cortex +
Insula
3a = Postcentral gyrus



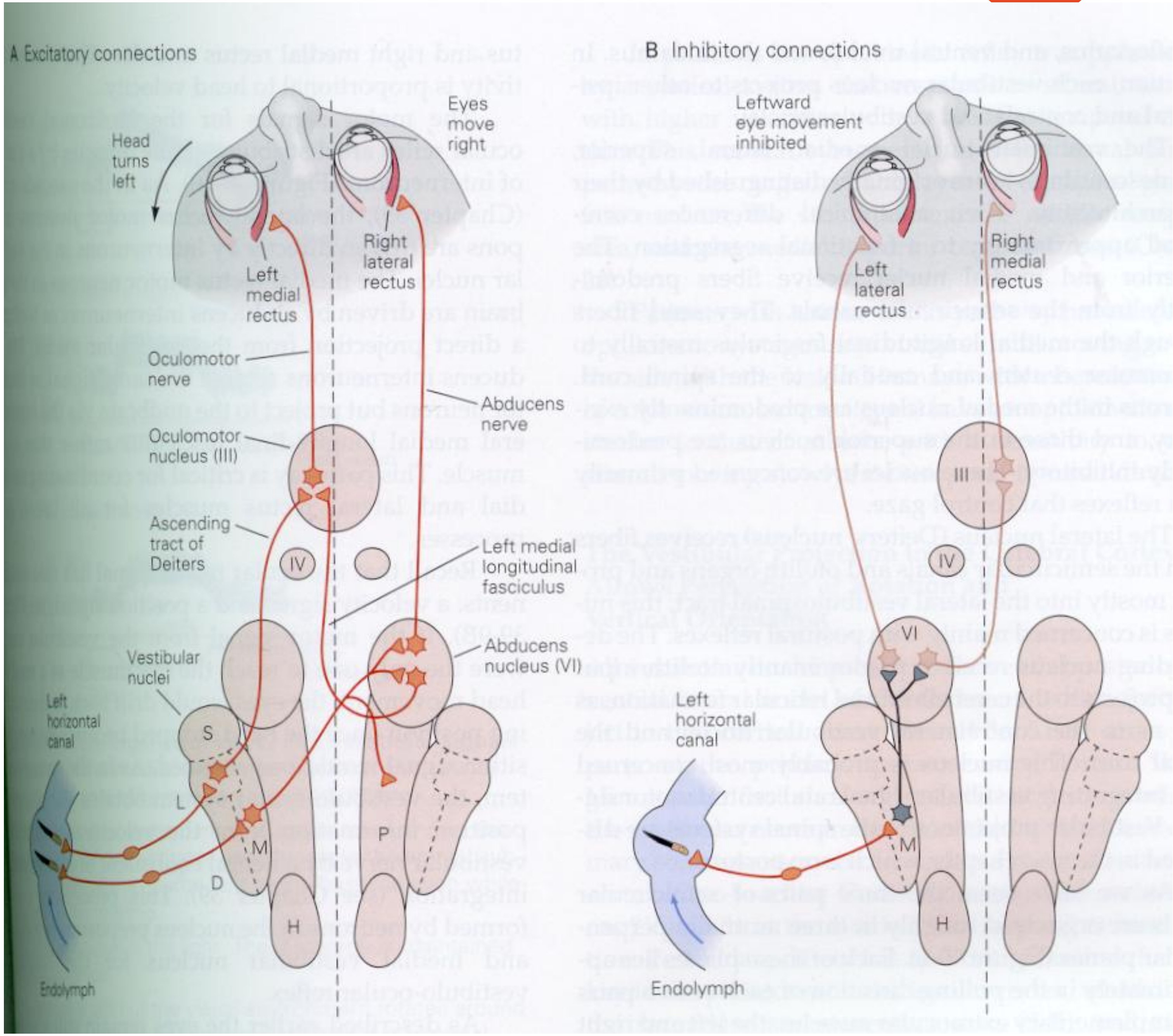
The Vestibulo-Ocular Reflex

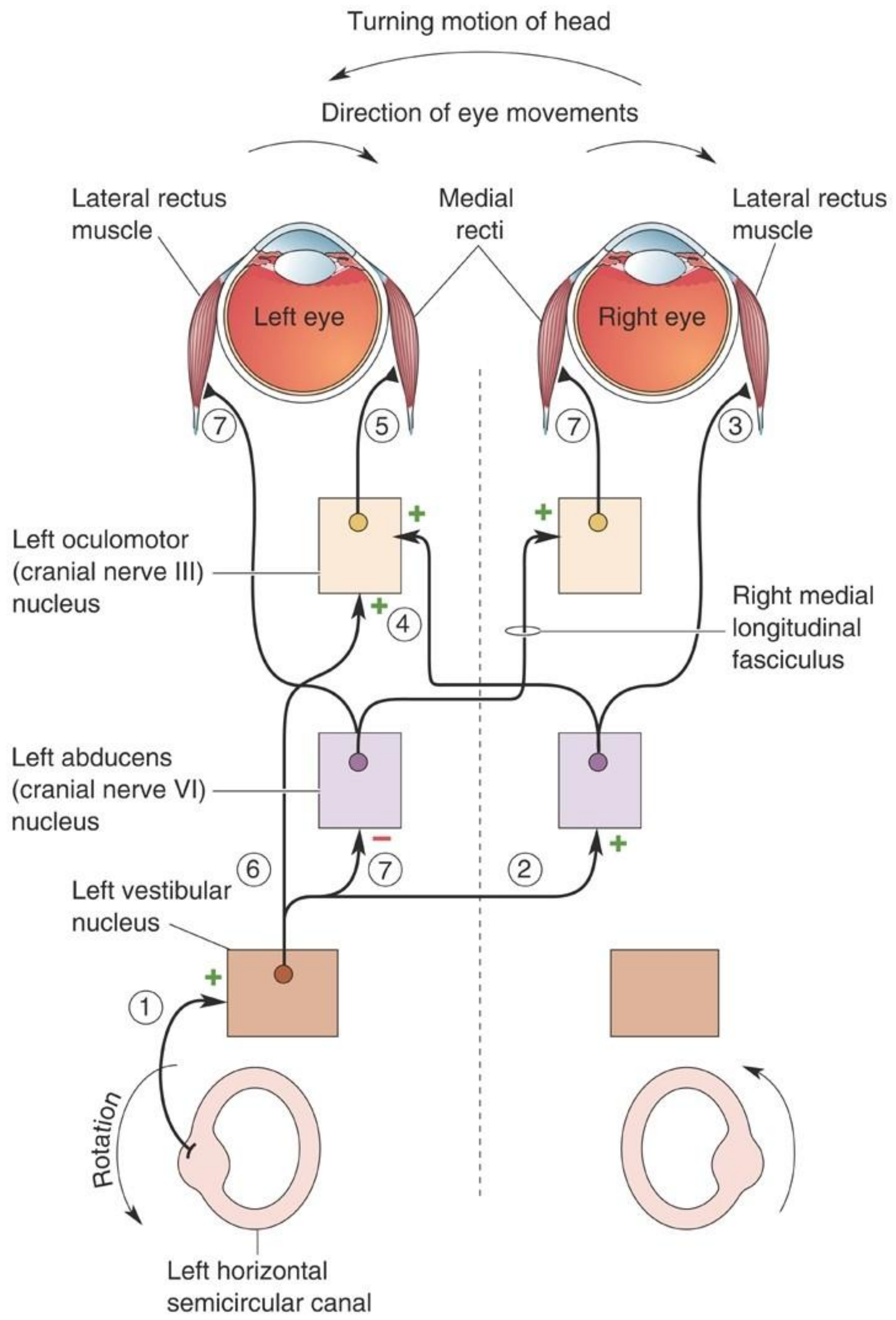
- As the head rotates toward one side, the eyes will initiate a slow reflexive movement to the contralateral side to keep the object of interest in the center of the visual field

OBJ. # 7

The Vestibulo-Ocular REFLEX

OBJ. # 7





Coordination of the VOR by the vestibular system triggered by head movement

OBJ. # 7

Clinical Correlations

Lesions of the vestibular pathways or the fibers of the inferior cerebellar peduncle generally produces:

- Pathological Nystagmus and
- Vertigo

Other symptoms and signs that could be associated are:

- Nausea and vomiting
- Hearing loss
- Wide-based gait
- Unsteadiness

Clinical Correlations

Pathological Nystagmus – Caloric Testing of vestibular function

* **Mnemonic**: COWS – cold=opposite, warm=same

Irrigation of the ear with cold water produces slow movements of the eyes to the ipsilateral side because convection currents mimic head movement away to the irrigated side

Irrigation of the ear with warm water produces slow movements of the eyes to the contralateral side because convection currents mimic head movement to the irrigated side